CIMS API INTEGRATION REQUIREMENTS

RAG-Based AI Chatbot for the Anti-Corruption Commission

Purpose: This document defines the CIMS information and API capabilities required to integrate CIMS with the ACC investigation chatbot. It is intended as a baseline for discussion and API design with the CIMS vendor. Endpoint paths and payloads below are proposed contracts, not claims about existing CIMS endpoints.

1. Executive Summary

The ACC investigation chatbot is being developed as a unified, natural-language interface for authorized ACC personnel to review investigation documents and correlate information across internal systems. The project has implemented core functionality for Retrieval-Augmented Generation, Call Data Record (CDR) analysis and Relationship Mapping.Integration with CIMS is the next major step in enabling the chatbot to analyze investigation documents alongside these existing sources.

The project team has selected authenticated API integration rather than direct live database access. CIMS remains the system of record; the chatbot will request only the information needed for an authorized query, process the returned information, and show results with traceable source references. The vendor is requested to confirm available data fields, access rules, authentication method, endpoint paths, response schemas, rate limits, and a test environment.

2. Objectives and Scope

- Retrieve authorized case and investigation metadata from CIMS.
- Retrieve case-linked documents and records, particularly witness statements, accused statements, FIRs, interview records, and investigation events, where available and permitted.
- Support summaries, entity extraction, timeline generation, cross-statement comparison, contradiction identification, and evidence correlation.
- Allow the chatbot to cite the originating CIMS case/document and, where possible, page or section references.
- Respect existing CIMS permissions and case assignments at the API/data-retrieval layer.
- Integrate CIMS information with existing CDR and Relationship Mapping outputs when the investigator's question requires cross-source analysis.

Out of scope for this integration: creating, editing, or deleting CIMS records; changing case assignments; bypassing CIMS authorization; and direct database access. The initial integration is read-only unless ACC separately approves additional operations.

3. Proposed Integration Architecture

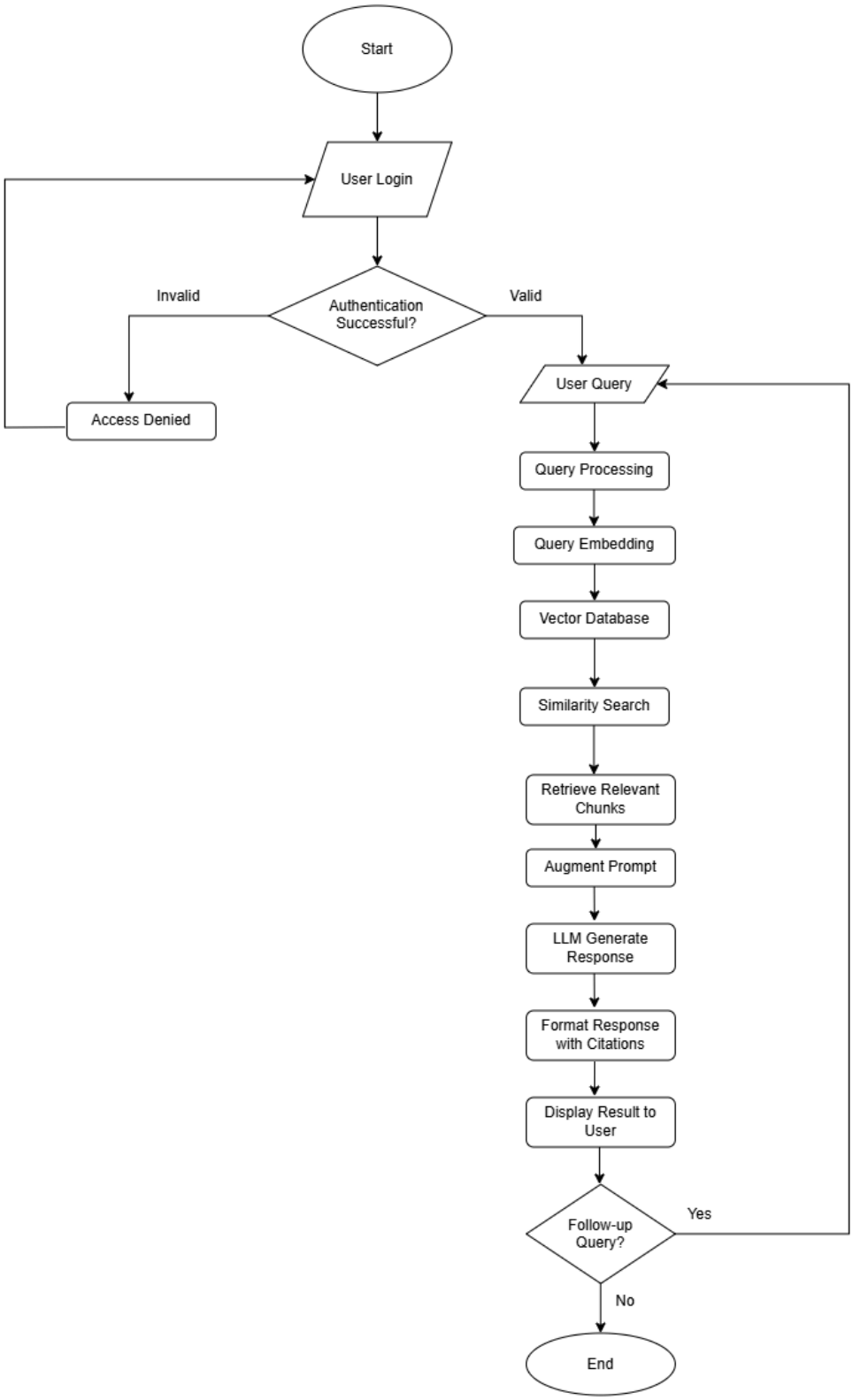
The chatbot backend will communicate with CIMS through a vendor-supported, authenticated REST API over the ACC-approved internal network. It will not connect directly to the CIMS

database. The API should return structured metadata and document content or a secure, short-lived document download link.

3.1 End-to-End Workflow

1. An ACC user signs in to the chatbot. The chatbot verifies the user identity and role.
2. The user submits a natural-language question, for example: “Summarize the witness statements for case X and compare the event dates.”
3. The backend identifies the likely intent and required source(s), and validates that the user may access the relevant case.
4. For CIMS-related requests, the backend retrieves the relevant case metadata, document listings, and document content through the vendor-provided APIs.
5. CIMS authenticates and authorizes the request, applies case-level permissions, and returns only permitted records.
6. The backend normalizes the returned data. Text documents may be cleaned and chunked; scanned/image-based documents may be passed through the chatbot’s OCR pipeline if text is not available.
7. For semantic document questions, the text chunks are embedded and indexed in the vector store with case, document, page, and permission metadata. For current structured case fields, the backend may use the API response directly rather than relying solely on vector search.
8. The retriever selects relevant authorized chunks/records. The backend assembles the evidence with the user’s question and sends it to the configured local language model.
9. The chatbot returns an evidence-grounded response with CIMS source references. Where useful, it may combine the result with CDR and Relationship Mapping API results.
10. The request, source identifiers, access outcome, and response metadata are logged for audit. AI-generated analysis is presented as decision support and not as a determination of wrongdoing.

3.2 RAG Flow


Start
User Login
Authentication Successful?
Invalid
Valid
Access Denied
User Query
Query Processing
Query Embedding
Vector Database
Similarity Search
Retrieve Relevant Chunks
Augment Prompt
LLM Generate Response
Format Response with Citations
Display Result to User
Follow-up Query?
Yes
No
End

CIMS API → Authorization validation → PDF retrieval → Text extraction/OCR → Text cleaning and chunking → Embedding generation → Qdrant vector storage → Authorized retrieval → Context assembly → Local language model → Source citations → Chatbot response

Important design note: vector-store filtering is an additional control, not a replacement for CIMS authorization. The integration must ensure that unauthorized records never enter the user's retrieval context. If access is revoked or case assignment changes, the chatbot must have a defined refresh/re-indexing or revalidation approach.

4. CIMS Information Required

The following table defines the information required for integration. The vendor should confirm which fields exist in CIMS, which are available through APIs, and which may be exposed to the chatbot under ACC's authorization policies.

Data category	Required fields / content	Why the chatbot needs it
Case / investigation metadata	Stable case ID; case/reference number; title or short description; status; case type/category; creation and update timestamps; assigned unit/team; authorized investigator/team identifiers where available.	Find cases, scope retrieval, identify current records, and apply permissions.
Document catalogue	Stable document ID; linked case ID; document type; title/name; upload/created/modified timestamps; status/version; MIME type; page count if available; confidentiality/classification; source URI or download mechanism.	List and identify relevant evidence without retrieving every file.
Document content	Extracted text or a secure way to retrieve the original file; page/section boundaries where available; language; document version; text extraction status.	Summarization, semantic retrieval, entity extraction, timeline generation, and source citations.

Investigation events	Event ID; case ID; event date/time; event type; description; linked person/entity/document IDs; created/updated timestamps.	Timeline generation and correlation with CDR events and statements.
People / entities linked to a case	Stable person/entity ID; display name; role in case (e.g., witness, accused, complainant); permitted identifiers; linked case/document IDs.	Resolve entities across records and support cross-document comparisons. Only fields authorized for the chatbot should be returned.
Access and audit context	Authenticated principal or service identity; permitted case scope or authorization decision; request/correlation ID; access outcome.	Enforce least privilege and maintain traceability.

The vendor should confirm which categories and fields actually exist in CIMS, which are exposed by API, and which are restricted. The list above is a requirements baseline, not an assertion that all fields exist in the current CIMS schema.

4.1. Synchronization Approach

Real-time processing of every document is not assumed to be a requirement for the initial implementation. Instead, the system will use scheduled synchronization, subject to ACC's operational requirements and the capabilities of the CIMS APIs.

The backend should check for newly added or modified documents at an agreed interval. Only affected documents should normally be reprocessed.

The synchronization interval will be determined after considering document volume, API availability, processing time, storage capacity, and the expected freshness of chatbot responses.

The system should support:

- Initial indexing of existing authorized documents.
- Incremental retrieval of new and modified documents.
- Identification of document versions to avoid unnecessary reprocessing.
- Handling of deleted, withdrawn, or access-restricted documents.

- Recovery from interrupted synchronization without silently skipping records.

The vendor should confirm whether CIMS supports an updated_since filter, change feed, webhook, or another suitable mechanism.

5. Proposed API Endpoints

The following REST paths are suggested for discussion. The vendor may retain existing conventions or propose equivalent paths, provided the same capabilities and security controls are supported. All endpoints are read-only and illustrative until agreed.

Method and proposed path	Purpose	Key inputs	Expected response
GET /api/cims/cases	Search/list cases visible to the authenticated user.	Optional case_number, status, updated_since, page, page_size.	Paginated case summaries; only cases the caller may access.
GET /api/cims/cases/{case_id}	Retrieve one case's authorized metadata.	case_id path parameter.	Case metadata and permitted linked-record counts/links.
GET /api/cims/cases/{case_id}/documents	List documents linked to a case.	case_id; optional document_type, updated_since, page, page_size.	Document IDs, types, timestamps, version/status and retrieval references.
GET /api/cims/documents/{document_id}	Retrieve document metadata and, where permitted, extracted text.	document_id.	Metadata plus text or a secure download reference; include page/section markers if available.
GET /api/cims/documents/{document_id}/content	Retrieve the document content through	document_id; optional version.	Text/plain or structured JSON content; alternatively

	an authorized API response.		vendor-approved secure download response.
GET /api/cims/me/permissions	Optional endpoint to inspect the authenticated integration identity's effective scopes.	Authenticated identity.	Permitted scopes/roles or a documented equivalent authorization mechanism.

5.1 Endpoint behaviour

The following behaviours are requested:

- Support pagination for list endpoints and deterministic sorting, preferably by updated_at and stable ID.
- Support incremental synchronization using updated_since or an equivalent change feed if approved for indexing.
- Return stable identifiers and timestamps so updates and deletions can be reconciled.
- Provide clear responses for missing records, denied access, and removed documents without disclosing restricted information.
- For large documents, prefer streamed or time-limited secure download rather than embedding large binary files in JSON.
- Expose only fields approved by ACC and necessary for the chatbot's functions.

5.2 Suggested response shapes

The following JSON examples are illustrative only. The vendor should provide the actual schemas, required fields, enumerated values, and date/time conventions.

Case list item (illustrative JSON):

```
{
  "case_id": "stable-case-id",
  "case_number": "ACC-CASE-REFERENCE",
  "title": "Case title or short description",
  "status": "OPEN",
  "updated_at": "2026-10-20T09:30:00Z"
}
```

Document metadata/content (illustrative JSON):

```
{
  "document_id": "stable-document-id",
```

```
  "case_id": "stable-case-id",
  "document_type": "WITNESS_STATEMENT",
  "title": "Document title",
  "version": "1",
  "created_at": "2026-10-18T10:00:00Z",
  "updated_at": "2026-10-20T09:30:00Z",
  "mime_type": "application/pdf",
  "page_count": 12,
  "text_content": "Extracted text if available",
  "page_text": [
    {"page": 1, "text": "Text for page one"}
  ],
  "source_reference": "vendor-approved source identifier"
}
```


These are examples only. The vendor should provide the actual schema, mandatory fields, enumerated values, date/time format, and document retrieval mechanism.

6. How CIMS Data Will Be Used

Chatbot capability	CIMS data needed	Processing / result
Case-specific Q&A	Case metadata, document catalogue and relevant content.	Retrieve evidence relevant to the question and answer with case/document citations.
Statement / FIR summarization	Statement/FIR/interview text or original file.	Clean or OCR text, summarize key facts, and retain source references.
Entity extraction	Relevant document text and available linked entity metadata.	Identify names, dates, locations, financial references and events for investigator review.
Timeline generation	Document text and structured investigation events.	Order dated events, preserve uncertainty, and cite the source for each event.
Contradiction / inconsistency review	Two or more authorized statements or records.	Compare claims and highlight apparent differences with quotations/page references;

		do not label a difference as proven misconduct.
CDR and case timeline correlation	CIMS case events/statements plus the existing CDR endpoints.	Compare reported events with communication timestamps and report possible matches or discrepancies.
Relationship and evidence correlation	Case-linked people/entities plus existing Relationship Mapping and CDR results.	Connect records by stable identifiers where possible and show the source for each link.

7. Current Implementation Progress

7.1 RAG and Document Processing

The core RAG pipeline has been implemented and tested.

Completed work includes:

- Document ingestion and text-processing functionality.
- Text chunking and embedding generation.
- Qdrant integration for vector storage and semantic retrieval.
- Retrieval of relevant investigation-document content.
- Language-model-based responses using retrieved context.
- Use of source metadata to associate retrieved content with its originating documents.

The next integration step is to connect the existing pipeline to the data retrieved from CIMS through the vendor-developed APIs.

7.2. CDR Analysis

Core CDR analysis APIs have been implemented.

The current functionality includes:

- Communication summaries and interaction statistics.
- Communication timeline retrieval.
- Date/time-filtered communication records.
- Detection of selected irregular communication patterns, including communication spikes, one-sided communication, short calls, and unusual communication hours.

- Integration of CDR functionality into the chatbot's retrieval and response workflow.

7.3. Relationship Mapping

Core relationship APIs have been implemented.

The current functionality includes:

- Retrieval of relationship information.
- Support for family relationships across the intended relationship degrees.
- Identification of connections between individuals.
- Integration of relationship information into chatbot query handling and response generation.

Further validation and integration will be required using the available ACC data and supported CIMS interfaces.

7.4. Evidence Correlation

Work is underway to combine information from investigation documents with structured CDR and relationship data.

The objective is to support cross-referencing of people, dates, events, communications, and relationships across available evidence sources.

Further development and testing will be required to validate this workflow using representative investigation cases and data retrieved from CIMS.

7.5. Existing Integrations and Endpoint References

The following endpoints document the existing project integrations. They are included to clarify how CIMS information will be combined with existing capabilities. The CIMS vendor is not expected to implement these endpoints unless separately

System	Endpoint documented by project	Purpose
CDR Analysis	GET /api/cdr/summary/{number}	Comprehensive communication summary, frequency, direction statistics, top contacts and data coverage.
CDR Analysis	GET /api/cdr/records/{number}	Filtered CDR records and statistics.

CDR Analysis	GET /api/cdr/timeline/{number}	Chronological communication timeline.
CDR Analysis	GET /api/cdr/patterns/{number}	Irregular communication pattern analysis.
Relationship Mapping	GET /api/relationship/{cid}/{degree}	Retrieve structured family relationships for a citizen; supported degrees 1, 2, and 3.

The report also mentions an earlier CDR route written as GET /api/summary/:number in the API design section. The implementation endpoint summary later lists /api/cdr/summary/{number}. Confirm the deployed route in the current environment before vendor-to-vendor testing.

8. Authentication, Authorization and Security Requirements

The integration must comply with ACC's security policies and the vendor's approved access-control model.

- Use HTTPS/TLS within the ACC-approved network. No sensitive case data should be sent to unapproved external services.
- Support a vendor-approved authentication mechanism such as OAuth 2.0 client credentials, mutual TLS, or signed service tokens. The vendor and ACC security team should select the mechanism; credentials must be stored securely and rotated.
- Use least-privilege, read-only access for the chatbot integration identity.
- Enforce authorization on every request at the CIMS API/data layer. The chatbot must not be able to retrieve a case merely by guessing its ID.
- Agree how the API represents the requesting ACC user and how case assignment/role restrictions are enforced. Prefer delegated user authorization or a trusted, verifiable user-context mechanism where supported.
- Return only the fields permitted for the caller and approved for the integration. Mask or omit sensitive identifiers unless explicitly needed and authorized.
- Log request/correlation ID, integration identity, user context where applicable, resource ID, timestamp, operation, and success/denial outcome. Avoid logging full statement text, credentials, tokens, or unnecessary personal data.
- Apply input validation, request-size limits, rate limits, timeouts, and safe error responses.
- Define data retention, caching, vector-index refresh, record deletion, access-revocation, and audit-review procedures before production use.

- Provide a non-production environment and synthetic or de-identified test records until written approval permits real case data.
- Document network allowlisting, certificates, secret rotation, incident contacts, API availability and maintenance windows.

9. Errors and Status Codes

HTTP status	Expected meaning
200 OK	Request succeeded; response follows the agreed schema.
400 Bad Request	Invalid parameter, date range, pagination value, or malformed request.
401 Unauthorized	Missing, invalid, or expired authentication.
403 Forbidden	Authenticated caller is not authorized for the requested case/document.
404 Not Found	Record does not exist or is not visible, following ACC-approved information-disclosure policy.
409 Conflict	Optional: version or synchronization conflict, if applicable.
429 Too Many Requests	Rate limit exceeded; include Retry-After if supported.
500 / 502 / 503	Server or upstream service failure; provide a correlation ID without exposing sensitive internals.

Error responses should use a consistent machine-readable structure, for example: {"error":{"code":"FORBIDDEN","message":"Access denied","request_id":"..."}}, subject to the vendor's established standards.

10. Operational and Data Synchronization Questions

- Are documents returned as extracted text, a secure file download, or both?
- Are scanned documents and handwritten records common, and does CIMS store OCR text or only the original files?
- How are amended documents, replaced versions, withdrawn records, and deleted records represented?
- Can the API expose updated_since filters, webhooks, or another approved change-notification mechanism?

- What are the request limits, maximum payload size, pagination rules, and expected response times?
- How should the chatbot respond when CIMS is unavailable or a document is still processing?
- Can the API return page-level text and stable source references so chatbot citations can open the original document in an authorized CIMS view?
- How are case assignments and permissions updated, and how quickly do changes take effect through the API?
- What test environment, sample schemas, OpenAPI/Swagger specification, and sample responses can the vendor provide?

11. Testing and Acceptance Criteria

- Authentication: valid credentials succeed; invalid, expired, or revoked credentials are rejected.
- Authorization: an authorized test user can retrieve assigned case records; a user without access cannot retrieve them, including by direct document ID.
- Data correctness: case metadata, document types, event timestamps and document content match approved CIMS test records.
- Pagination and filtering: list endpoints return consistent pages and correctly apply case, type, and update-time filters.
- Document traceability: every returned document has a stable ID and enough metadata to cite its case, document type and page/section when available.
- RAG grounding: answers based on CIMS documents include source references and do not invent facts absent from retrieved evidence.
- Cross-source analysis: a test query can combine CIMS events with CDR timeline/pattern output or relationship data when authorized.
- Error handling: missing records, denied access, empty results, rate limits, timeouts and upstream failures are handled safely.
- Auditability: requests and access outcomes are traceable through correlation IDs without unnecessarily copying sensitive content into logs.
- Revocation/update: changes to case access or source documents are reflected within an agreed synchronization/revalidation window.

12. Vendor Deliverables Requested

1. Confirmation of available CIMS API capabilities and the proposed endpoint paths.
2. OpenAPI/Swagger specification or equivalent API documentation.
3. Field-level response schemas, data dictionaries, enumerated document types/statuses and example responses.
4. Authentication and authorization specification, including how case-level permissions are enforced.

5. Non-production API base URL, network requirements, test credentials/certificates and synthetic test records.
6. Document-content retrieval approach, including page-level text/source references and secure file access if applicable.
7. Pagination, filtering, rate-limit, timeout, versioning and error-handling details.
8. Change tracking/deletion strategy and guidance for handling updated documents and access revocation.
9. Named technical contact and process for reporting integration defects or security concerns.

13. Decisions to Confirm with ACC and Vendor

Decision	**Proposed baseline**
Integration method	Authenticated REST API; no direct live database access.
Permissions	CIMS remains authoritative; enforce case/document authorization on every request.
Initial operation	Read-only retrieval. Write operations are excluded.
Document delivery	Structured metadata plus extracted text where available; otherwise secure file retrieval.
Indexing	Index only approved data; retain case/document/page/source IDs and access metadata. Agree refresh/revalidation and deletion behaviour.
Deployment	ACC-approved secured/on-premises environment; no unapproved external transmission.
API contract	Vendor may propose existing endpoints equivalent to the proposed paths in Section 5.

14. Conclusion

The requested CIMS API will allow the ACC investigation chatbot to retrieve authorized case metadata, investigation documents and events, and use them in a source-grounded RAG workflow. Combined with the existing CDR and Relationship Mapping integrations, this will support document analysis and cross-source investigation assistance while preserving CIMS as the system of record. The proposed endpoints and schemas are a starting point for technical discussion; the final contract should be agreed by the CIMS vendor, ACC technical/security representatives, and the project team.